

Biology 0610

Course notes

Sample book for the Klaro study view. Written for the demo;
not an exam board text.

1 Cells

1.1 Cell structure

Parts every cell has

Every living cell has three parts in common. The cell membrane holds the cell together and controls what enters and leaves it. The cytoplasm is the jelly where most chemical reactions happen, and it holds the mitochondria, where aerobic respiration releases energy, and the ribosomes, where proteins are made. The genetic material, DNA, is kept inside the nucleus in animal and plant cells. When you label a diagram, draw a straight label line that touches the part and never cross two lines.

The cell membrane

The membrane is partially permeable: small molecules such as water, oxygen and carbon dioxide pass through it, while large molecules such as starch cannot. This is what lets the cell keep useful substances in and waste out. Do not confuse the membrane with the cell wall. The wall is stiff and fully permeable, so it gives shape and support but controls nothing. Examiners often ask which part controls what enters and leaves the cell: the answer is always the membrane.

Parts only plant cells have

A plant cell has three parts an animal cell does not. The cellulose cell wall supports the cell and stops it bursting when it takes in water. The permanent vacuole is filled with cell sap and pushes the cytoplasm against the wall, which keeps the cell firm. Chloroplasts contain chlorophyll, which absorbs light energy for photosynthesis. Not every plant cell has chloroplasts: root cells live in the dark and have none, so never write that all plant cells photosynthesise.

1.2 Specialised cells

Red blood cells

A red blood cell carries oxygen from the lungs to the tissues. It has no

nucleus, which leaves more room inside for haemoglobin, the red pigment that joins with oxygen. Its biconcave disc shape gives a large surface area for oxygen to diffuse in and out quickly, and it is small and flexible enough to squeeze through narrow capillaries. In an answer, always link the feature to what it does: no nucleus, so more haemoglobin, so more oxygen carried.

Root hair cells

A root hair cell absorbs water and mineral ions from the soil. Its long thin extension pushes between soil particles and greatly increases the surface area for absorption. Water enters by osmosis because the cell sap is more concentrated than the soil water. Mineral ions such as nitrate are often more concentrated inside the cell than in the soil, so they must be taken in by active transport, which uses energy from respiration. That is why root hair cells contain many mitochondria.

2 Movement in and out of cells

2.1 Diffusion

What diffusion is

Diffusion is the net movement of particles from a region of higher concentration to a region of lower concentration, down a concentration gradient, as a result of their random movement. It needs no energy from the cell: the kinetic energy of the particles is enough. Oxygen diffuses into the blood in the alveoli, and carbon dioxide diffuses into a leaf through the stomata. The word net matters, because particles still move both ways; more of them simply move down the gradient.

What changes the rate

Four things change how fast diffusion happens. A larger surface area gives more room for particles to cross. A steeper concentration gradient makes the net movement faster. A higher temperature gives the particles more kinetic energy. A shorter distance, such as a thin membrane or a wall one cell thick, means particles arrive sooner. When a question asks you to explain an adaptation, name the factor it changes and say that it makes diffusion faster.

2.2 Osmosis and active transport

Osmosis

Osmosis is the net movement of water molecules from a region of higher water potential, a dilute solution, to a region of lower water potential, a concentrated solution, through a partially permeable membrane. An animal cell placed in pure water takes in water until it bursts, because it has no wall. A plant cell swells until the wall pushes back and it becomes turgid. In a concentrated solution a plant cell loses water, becomes flaccid, and its membrane may pull away from the wall: the cell is plasmolysed.

The potato strip experiment

Cut equal strips of potato, blot them dry and record the mass of each. Place one strip in each of a range of sugar solutions and leave them for thirty minutes at the same temperature. Blot again, reweigh, and work out the percentage change in mass so that strips of different starting mass can be compared. Strips that gain mass were in a solution with a higher water potential than the potato cells; strips that lose mass were in a lower one. Where the line of your graph crosses zero, the solution has the same water potential as the cells.

Active transport

Active transport is the movement of particles through a cell membrane from a region of lower concentration to a region of higher concentration, against a concentration gradient, using energy from respiration. Carrier proteins in the membrane pick up the particle on one side and release it on the other. It lets root hair cells take in mineral ions from dilute soil water and lets the small intestine absorb glucose that is already at a low concentration in the gut. If respiration stops, active transport stops with it.